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**Figure 1** A pasture in a meadow, with woolly and sheared sheep, and grass and dirt blocks.


# Population Models of *Minecraft* Sheep

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## Introduction

*Minecraft* is an open-world sandbox video game*. The game takes place in a procedurally generated world made of cubic voxels, or *blocks*. Among other activities, players can engage in agriculture, including the raising of sheep.

Players can gather blocks of wool by raising a flock of sheep in a pasture (Figure 1). When a sheep is sheared, they must eat grass to regrow their wool. When a grass block is eaten, it converts into dirt, and can regrow if grass spreads from an adjacent grass block.

One may be tempted to pack the pasture as densely as possible to maximize space efficiency. However, if a pasture is packed too densely with sheep, they will deplete the grass, and wool production will be low. On the other hand, if the pasture is not densely packed, wool production will still be low because there are not many sheep. A mathematical population model can give insight into the happy medium sheep population.

*This investigation was conducted in *Minecraft* Java edition version 1.19.4.

## Table of symbols

| | |
|---|---|
| $N$ | Sheep population |
| $A$ | Pasture size |
| $N_g$ | Grass population |
| $\lambda$ | Grass growth rate |
| $\mu$ | Grass consumption rate |
| $k$ | Grass growth rate constant |
| $R$ | Grass consumption rate constant |
| $P$ | Pasture carrying capacity |
| $p_x$ | Probability distribution of $N_g$ |
| $q_{xy}$ | Probability distribution of $N_g$ and $N_w$ |

## Deterministic model

**Grass population** The *Minecraft* physics engine operates in discrete time increments of 0.05 seconds, called *ticks*. Here, this will be approximated as a continuous-time process.

Sheep randomly attempt to eat the block they're standing on. If the sheep and grass in the pasture are uniformly distributed in a pasture with $A$ total blocks, $N_g$ grass blocks, and $N$ sheep, the grass consumption rate $\mu$ is given by:

$$\mu = \frac{NRN_g}{A} \tag{1}$$

Where $R$ is the frequency with which an individual sheep attempts to eat.

Grass blocks randomly spread to adjacent dirt blocks. If $N_g$ is small, then grass growth will be limited by the lack of places the grass can spread *from*. If $N_g$ is large, grass growth will be limited by a lack of places for the grass to spread *to*. A simple approach is to assume the growth rate $\lambda$ has a parabolic dependence on $N_g$, such that $\lambda = 0$ for $N_g = 0$ and $N_g = A$:

$$\lambda = kN_g - \frac{k}{A}N_g^2 \tag{2}$$

Where $k$ is a constant related to the rate at which grass spreads.

The overall rate of change of $N_g$ is the difference of $\lambda$ and $\mu$:

$$\dot{N}_g = \lambda - \mu = \left(k - \frac{NR}{A}\right) N_g - \frac{k}{A}N_g^2 \tag{3}$$

This is the *logistic* model [**1**], one of the earliest and simplest models for population growth. It is seldom employed for real-life ecology, having been supplanted by more intricate and flexible models [**2**], but it is applicable here due to the simple mechanics of *Minecraft*.

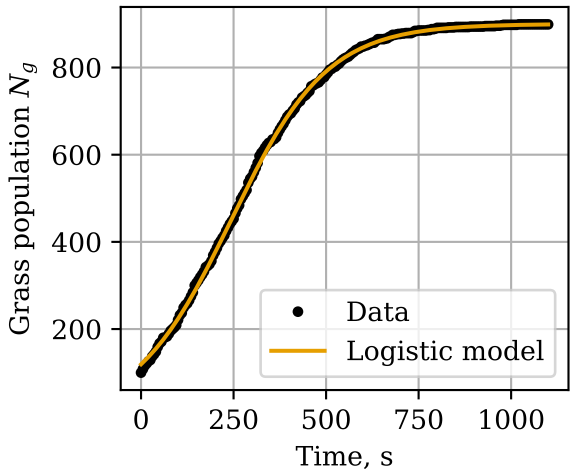
The exact solution [**2**] of equation 3 is:

$$N_g(t) = \frac{P}{1 + \left(\frac{P}{N_g(0)} - 1\right) e^{-\left(k - \frac{NR}{A}\right)t}} \tag{4}$$

Where

$$P = A - \frac{NR}{k}$$

**Figure 2** Logistic model fit to grass growth data.




The population asymptotically approaches $P$ as $t \to \infty$. In ecology, $P$ is known as the carrying capacity, the maximum population that a habitat can sustain in equilibrium. This is not to be confused with $A$, the number of blocks in the pasture and therefore the maximum number of grass blocks it can contain. Note also that for $N \geq \frac{Ak}{R}$, the carrying capacity is 0.

**Experimental determination of rate constants** The actual values of $k$ and $R$ were determined with in-game experiments. $R$ was measured by counting the number of grass blocks eaten by sheep. In 1 hour, 10 sheep ate 320 blocks of grass, which gives $R = 0.00888\ \mathrm{s}^{-1}$.

$k$ was measured by uniformly dispersing 100 grass blocks in a 900-block dirt patch and tracking the grass population over time. The grass growth was then fit to equation 4. This fit gave $k = 0.00772\ \mathrm{s}^{-1}$. More details about these experiments can be found on the author's Github*.

**Woolly sheep population** A naïve assumption is that the farmer has superhuman speed, and shears each sheep the instant it grows its wool. In this case, the rate at which wool is gathered† is equal to $\mu$. Denoting the rate of wool gathering with $F$, and substituting $N_g = P$ into equation 1 (assuming the pasture has equilibrated):

$$F_{fast} = \mu = NR - \frac{N^2R^2}{Ak} \tag{5}$$

This is a simple parabolic relationship with a maximum value of $F_{fast} = \frac{Ak}{4}$ at $N = \frac{Ak}{2R}$.

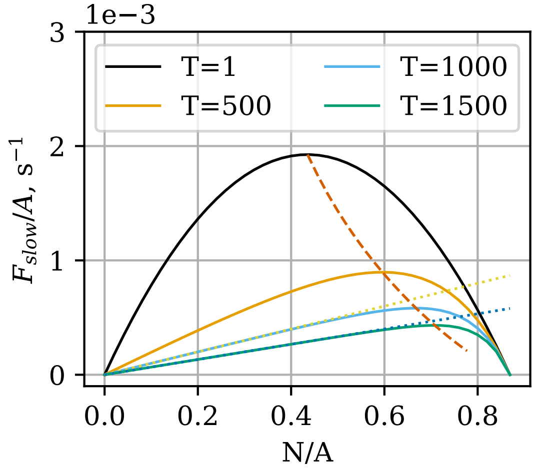
A more realistic assumption is that the farmer periodically shears all the sheep at once. Let $N_w(t)$ be the number of woolly sheep at time $T$. If the farmer shears the sheep with period $T$, the average rate at which they gather wool is:

$$F_{slow} = \frac{N_w(T)}{T}$$

*github.com/ELFREELAND

†Shearing a sheep actually yields between 1-3 blocks of wool, but this is out of the player's control, so it is assumed here that a sheep always yields one wool.

**Figure 3** Contours of $F_{slow}$ as a function of N at constant T. The dashed line traces the optimal $N$ and $F_{slow}$ as $T$ increases.




Analyzing this model requires analyzing the evolution of $N_w$ with time. A sheep will only regrow wool upon eating if it doesn't already have any, so one should expect that the rate at which sheep regrow wool is proportional to the number that are woolless:

$$\dot{N}_w = \frac{N - N_w}{N}\mu = (N - N_w)\frac{RN_g}{A} \tag{6}$$

Solving equation 6 with the condition $N_w(0) = 0$, and assuming $N_g = P$ (and is therefore time-independent) yields:

$$N_w(T) = N - Ne^{-TR + \frac{TR^2N}{Ak}}$$

Therefore:

$$F_{slow} = \frac{N - Ne^{-TR + \frac{TR^2N}{Ak}}}{T} \tag{7}$$

The optimal $N$ can be found by differentiating equation 7 w.r.t. $N$, setting $\frac{dF_{slow}}{dN} = 0$, and solving for $N$:

$$N = \frac{Ak}{TR^2}(W(e^{TR+1}) - 1) \tag{8}$$

Where $W$ is the Lambert $W$ function, which satisfies $W(xe^x) = x$. Plugging this back into equation 7 gives the value of $F_{slow}$ at optimal $N$:

$$F_{slow} = \frac{Ak(W(e^{TR+1}) - 1)(1 - e^{-TR + W(e^{TR+1}) - 1})}{R^2T^2} \tag{9}$$

Figure 3 shows a plot of equations 7-9. As $T$ increases, the optimal sheep population also increases, and the optimal rate of wool gathering decreases. Clearly there is a tradeoff between the frequency with which the farmer shears the flock and the rate at which they gather wool.

For small $T$, $F_{slow} \approx F_{fast}$. This can be shown quantitatively by employing a first-order approximation for $e^x$, for $x \approx 0$: $e^x \approx x + 1$. In equation 7, the exponent is small for small values of $T$, therefore:

$$F_{slow} \approx \frac{N - N(-TR + \frac{TR^2N}{Ak} + 1)}{T} = NR - \frac{N^2R^2}{Ak} = F_{fast}$$

It appears in Figure 3 that for large $T$ and small $N$, $F_{slow}$ is approximately linear w.r.t. $N$. Under these conditions, all of the sheep have regrown their wool by the time the farmer comes back, and $N_w(T) \approx N$. Lines of $\frac{N}{T}$ are plotted over top of the curves in Figure 3.

## Stochastic model

**Grass population** A deterministic model considers population as a single number. A stochastic model considers the *probability distribution* of the population. Here, the grass and sheep will be modelled with a *Markov chain*. The derivation of this method is described in [**3**], and applied here for the case of the logistic model.

Let $p_x(t)$ be the probability that the pasture has $N_g = x$ at time $t$. The time derivative of $p_x$ is given by the differential equation:

$$\dot{p}_x(t) = -p_x(t)(\lambda(x) + \mu(x)) + p_{x-1}(t)\lambda(x-1) + p_{x+1}(t)\mu(x+1) \tag{10}$$

The first term represents the evolution of the pasture away from $N_g = x$ due to growth or consumption, the second and third terms represent the evolution of the pasture toward $N_g = x$ by growth and consumption, respectively.

Note that $\mu(x)$ simply represnts the value of $\mu$ with $N_g = x$, and likewise for $\lambda$. A pasture of size $A$ will have a system of $A + 1$ equations, one for each $x$. Systems of equations such as these, which describe the evolution of a probability distribution with time, are known as *Kolmogorov equations* [**3**]. In general, Kolmogorov equations can be used to describe a *Markov process*, i.e. a process by which a system transitions between multiple possible states. In this case the states are the possible values of $N_g$.

In vector form, this system is expressed as $\dot{\mathbf{p}}(t) = \mathbf{p}(t)\mathbf{R}$, where $\mathbf{p}(t) = [p_0(t), p_1(t), p_2(t) \ldots p_A(t)]$, and $\mathbf{R}$ is a matrix of coefficients, such that the coefficient $r_{i,j}$ relates $p_i$ with $\dot{p}_j$:

$$\dot{p}_j = \sum_i r_{ij}p_i$$

$$r_{i,j} = \begin{cases} \lambda(j-1) & \text{for } i = j-1 \\ \mu(j+1) & \text{for } i = j+1 \\ -(\lambda(i) + \mu(i)) & \text{for } i = j \\ 0 & \text{otherwise} \end{cases}$$

This system was solved numerically for pastures of sizes $A = 100$, $200$, and $500$, and a range of $N$ values. *Minecraft* pastures typically begin fully covered with grass, so the initial condition will be $\mathbf{p}(0) = [0, 0, 0 \ldots 1]$.

Figure 4 compares the expected value of $N_g$ after 10000 seconds with the deterministic carrying capacity. They line up very closely for small sheep densities, but for large sheep densities, the expected $N_g$ is significantly less than $P$, especially for $A = 100$.

Figure 5 shows the evolution of the distribution of $N_g$ for $N = 40$ and $N = 70$ in a 100-block pasture. The $N = 40$ pasture stabilizes, but the $N = 70$ pasture continually evolves toward lower and lower $N_g$, with $p_0$ increasing dramatically as time proceeds.

**Figure 4** Expected value of $N_g$ at $t = 10000$, for $A = 100$ and $A = 500$. $P$ from the deterministic model is shown as a dotted line.

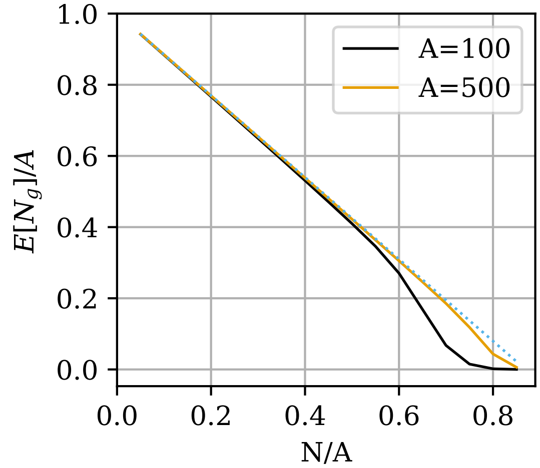
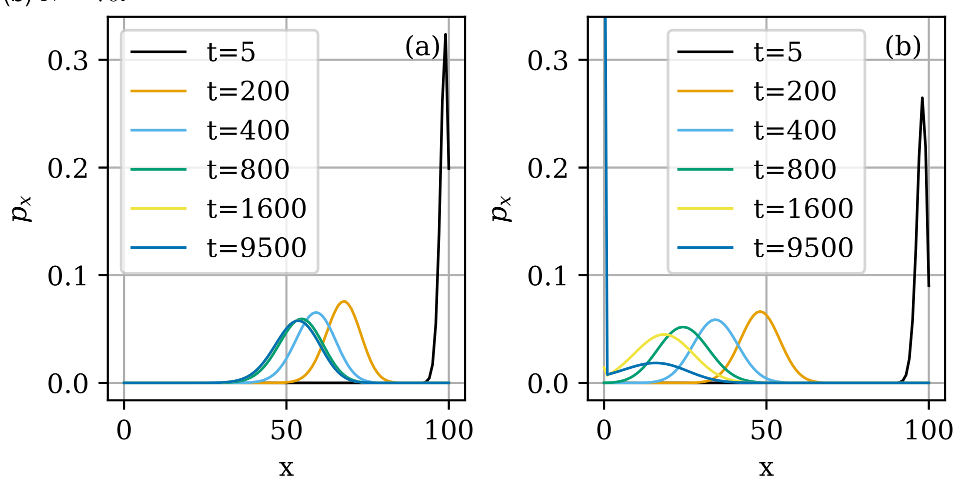



**Figure 5** Evolution of $N_g$ distribution for a pasture of size $A = 100$, for (a) $N = 40$ and (b) $N = 70$.




The reason for this is evident from the equation for $\dot{p}_0$. Beginning with equation 10 and considering that $p_{-1} = 0$ and $\lambda(0) = \mu(0) = 0$:

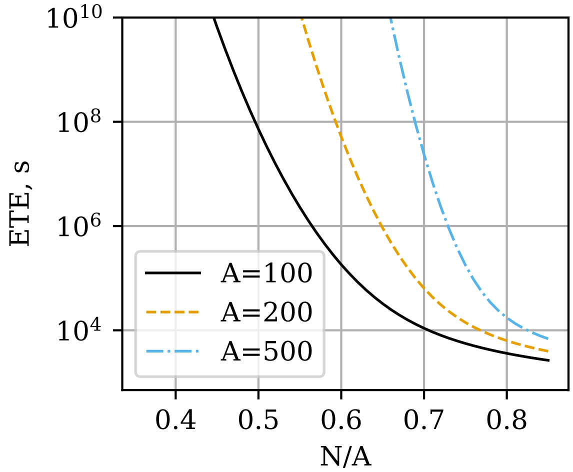
$$\dot{p}_0(t) = p_1(t)\mu(1)$$

Since $p_0$ is monotonically increasing, the true equilibrium state is $\mathbf{p} = [1, 0, 0, 0, \dots]$ i.e. extinction. However, even if this is the equilibrium state, the pasture may remain at a different *metastable* (or *quasi-equilibrium*) distribution if the probability of extinction remains negligible across reasonable time scales, as seen in Figure 5a.

Rapid grass extinction at high sheep densities poses a problem, since the deterministic model predicts that the optimal $N$ increases as $T$ increases (Figure 3). It can be shown [**4**] (see also [**3**] section 6.8.1.2.) that taking the negative inverse of the coefficient matrix for transient (i.e., nonzero) states, and summing across row $x$ gives the expected extinction time for a starting population of $x$.

Figure 6 shows the expected extinction time as a function of $N$. For all three pasture

**Figure 6** Expected time to extinction for pastures of $A = 100$, $200$, and $500$ as a function of sheep population density.




sizes, populations close to $\frac{Ak}{R} = 0.869$ have extinction time on the order of a few hours. The extinction time increases dramatically with decreasing sheep density and increasing pasture size, with extinction time on the order of several years for sheep densities less than $0.7$ in a 500-block pasture.

**Woolly sheep population** A model for both the sheep and grass mechanics will involve a bivariate probability distribution, $q_{xy}$, where:

$$q_{xy}(t) = \Pr(N_g(t) = x, N_w(t) = y)$$

Following the same prodecure as with equation 10:

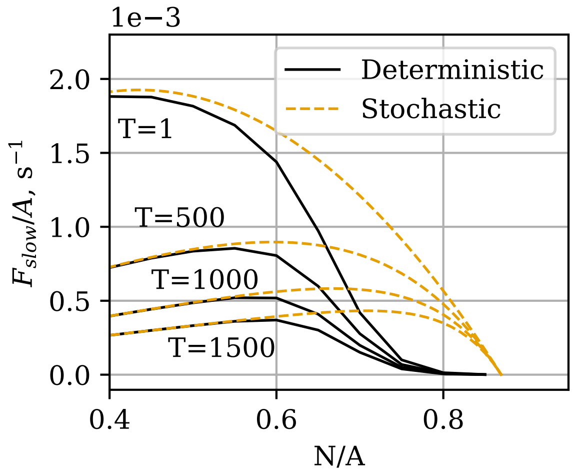
$$\begin{aligned} \dot{q}_{x,y}(t) &= q_{x,y}(t)(-\mu(x) - \lambda(x)) \\ &\quad + q_{x-1,y}(t)\lambda(x-1) \\ &\quad + q_{x+1,y}(t)\mu(x+1)\frac{y}{N} \\ &\quad + q_{x+1,y-1}(t)\mu(x+1)\frac{N-(y-1)}{N} \end{aligned} \tag{11}$$

The resulting system of $(A+1)(N+1)$ equations can be expressed as $\dot{\mathbf{q}}(t) = \mathbf{q}(t)\mathbf{R}$, where $\mathbf{q}(t)$ is a matrix of elements $q_{ij} = q_{xy}(t)$, and $\mathbf{R}$ is a rank-4 tensor:

$$r_{ijkl} = \begin{cases} -\mu(i) - \lambda(i) & \text{for } (i,j) = (k,l) \\ \lambda(k-1) & \text{for } (i,j) = (k-1,l) \\ \mu(k+1)\frac{l}{N} & \text{for } (i,j) = (k+1,l) \\ \mu(k+1)\frac{N-(l-1)}{N} & \text{for } (i,j) = (k+1,l-1) \\ 0 & \text{otherwise} \end{cases}$$

As before, this system can be solved numerically. The initial values of $q_{x,0}$ were set using the values of $p_x(t = 10000)$ from the grass population model. 7 shows $F_{slow}$ as calculated from the results of this model. Similarly to Figure 4, the stochastic model predicts significantly lower $F_{slow}$ than the deterministic model.

**Figure 7** $F_{slow}$ calculated from the stochastic model for sheep and grass, for a pasture with $A = 100$. The initial state was obtained from the numerical solution of $p_x$ at $t = 10000$.




The root cause of this deviation is the same as for the grass growth model. The equations for $\dot{q}_{0,y}$ are:

$$\dot{q}_{0,y}(t) = q_{1,y}(t)\mu(1)\frac{y}{N} + q_{1,y-1}(t)\mu(1)\frac{N-(y-1)}{N}$$

Since all terms here are positive, $q_{0,y}$ are monotonically increasing, and given enough time, the pasture will tend to evolve into one of these states and never leave.

The biggest drawback of this model is its computational intensiveness. A model for a pasture with $A = 500$ and density 0.8 would have $(500+1)(400+1) = 200901$ simultaneous equations, and an $\mathbf{R}$ tensor with $200901^2 = 40361211801$ elements, which would occupy 161 GB of memory if stored as 64-bit floats, making computation infeasable for all but top-shelf consumer hardware.

## Conclusions

The logistic model, and the Kolmogorov equations, have proven useful for predicting the behavior of a pasture in *Minecraft*. Future work includes investigating more realistic relationships for grass growth, especially adding an ”immigration” term to account for grass spreading into the pasture from outside, and assessing the effect of a continuous-time model for a discrete-time process.

## REFERENCES

1. Verhulst, P. “Notice sur la loi que la population suit dans son accroissement.” *Correspondance Mathématique et Physique* 10 (1838): 113-21.
2. Banks, Robert B. *Growth and Diffusion Phenomena*. Texts in Applied Mathematics. Springer Berlin, Heidelberg, 1994.
3. Matis, James, and Thomas Kiffe, eds. *Stochastic Population Models: A Compartmental Perspective*. 1st ed. Lecture Notes in Statistics. New York: Springer, n.d.
4. Matis, J. H., T. E. Wehrly, and C. M. Metzler. “On Some Stochastic Formulations and Related Statistical Moments of Pharmacokinetic Models.” *Journal of Pharmacokinetics and Biopharmaceutics* 11, no. 1 (1983): 77-92. https://doi.org/10.1007/BF01061769.